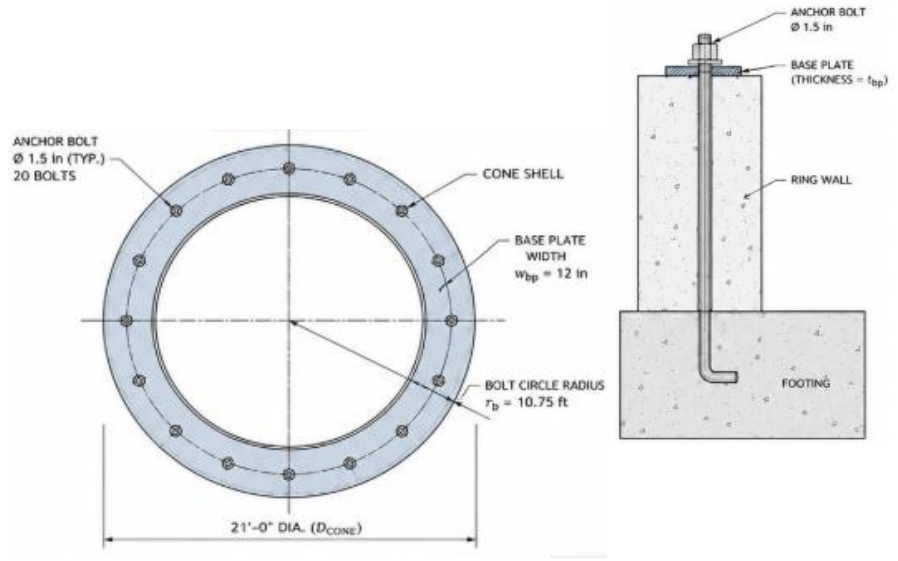
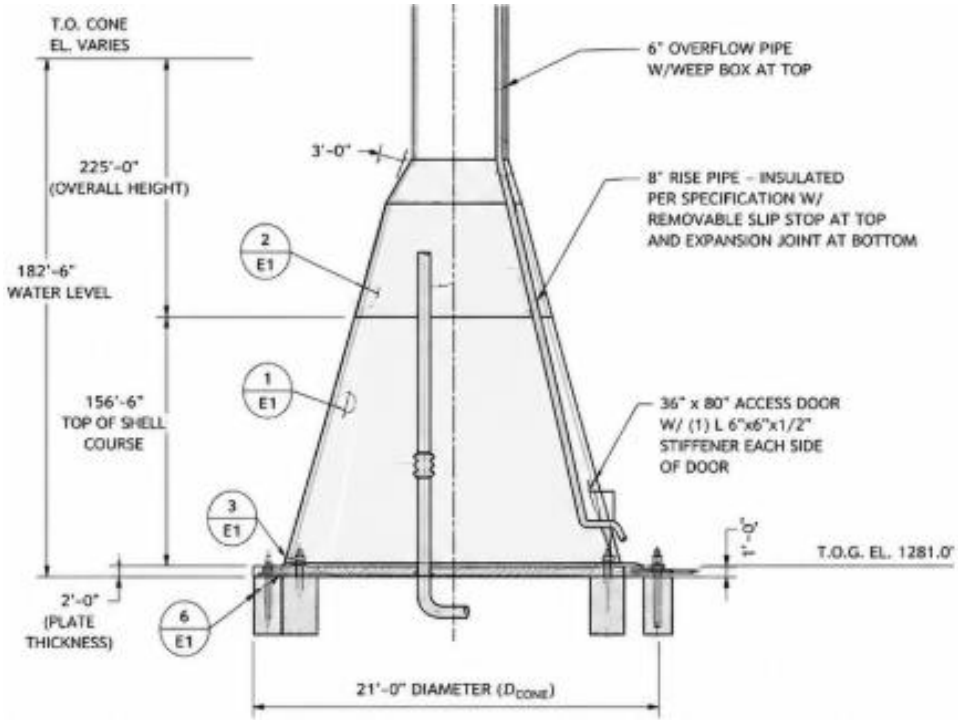
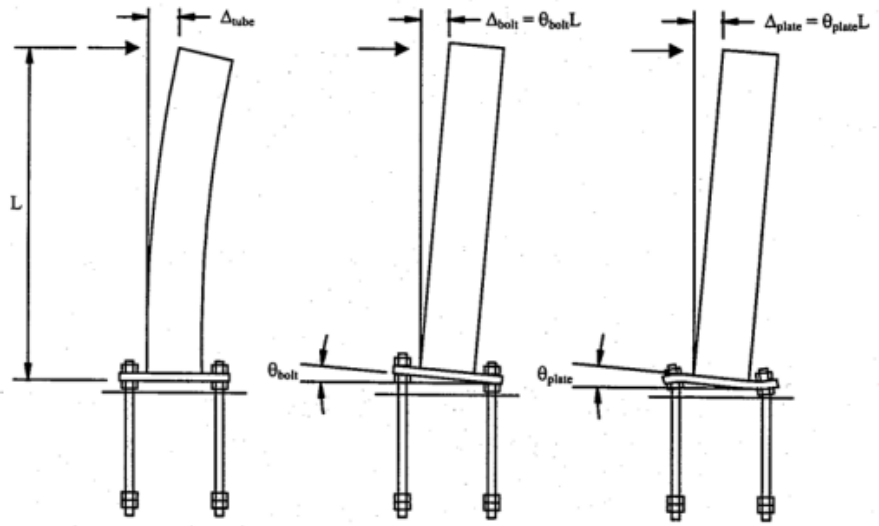
2. Base plate segment

Parameter	Symbol	Value	Type
Cone diameter	Dcone	21’	PREDEFINED
Width of ring wall	Wrw	2’	PREDEFINED
Number of anchor bolts	Nb	20	PREDEFINED / inherited from Anchor Bolt module
Anchor bolt diameter	db	1.5 in	PREDEFINED / Anchor Bolt module
Design Strip	N	12 in	PREDEFINED
Vertical load on foundation	P	1650.2553 kip	PREDEFINED / Main Tank Design
Governing overturning moment	Mu	2903.3789 kip-ft	PREDEFINED / Main Tank Design
Steel yield strength	Fy	36 ksi	USER INPUT
Concrete compressive strength	f'c	4 ksi	USER INPUT
Steel unit weight	γ_s	490 pcf	PREDEFINED
Number of plate segments	Ns	4	PREDEFINED
LRFD resistance factor	Ø	0.90	PREDEFINED
Modulus of elasticity	E	29,000 ksi	PREDEFINED





1.1. Geometry Inputs Derived from These:



Outer Radius

$R_o = \frac{D_{cone}}{2} + \frac{w_{rw}}{4}$

$\boldsymbol{R_o = 11ft}$

Inner Radius

$R_i = \frac{D_{cone}}{2} - \frac{w_{rw}}{4}$

$\boldsymbol{R_i = 10ft}$

2.0. Design Against Moment:

Based on the AISC LRFD moment design procedure adapted for annular strip behavior.

2.1. Determine Factored Load and Moment

Given

$P_u = 1650.2553$ kips

$M_u = 4847.67$ kip-ft

Convert moment to kip-in:

$$M_u = 2903.3789(12) = \mathbf{34840.55\ kip\text{-}in}$$

2.2. Determine Circumferential Moment per Unit Strip

For annular strip design, distribute moment around the circumference.

Governing Equation

$$M_{strip} = \frac{M_u}{\pi D_{cone}}$$

Substitute:

$$M_{strip} = \frac{34840.55}{\pi(21)}$$

$$M_{strip} = \frac{34840.55}{65.973}$$

$M_{strip} = 528.10$ kip-in/ft

Convert to kip-in/in:

$$M_{strip} = \frac{528.10}{12}$$

$\boldsymbol{M_{strip} = 44.0083}$ **kip-in/in**

2.3. Determine the maximum design bearing stress

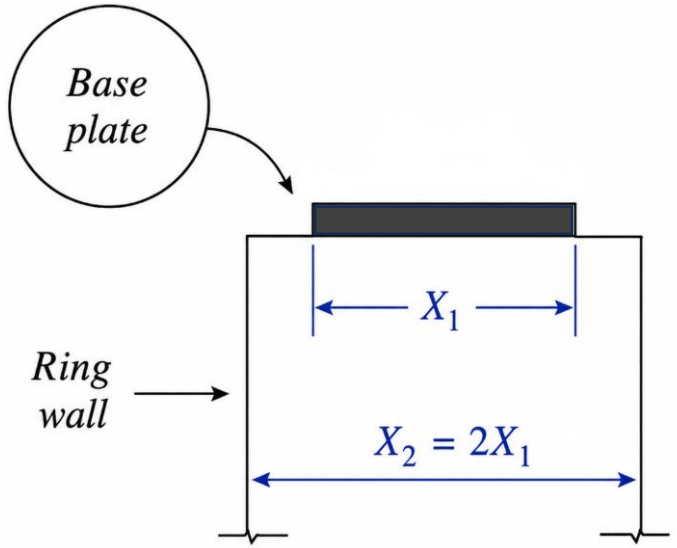
$$\frac{A_2}{A_1} = \frac{x_2\,(rw)}{x_1\,(wbp)} = \frac{x_2}{x_1} = \frac{2}{1} = 2$$

$$F_p = 0.85\phi f_c' \sqrt{\frac{A_2}{A_1}} \leq 1.7 f_c'$$

$$0.85 * 0.9 * 4 * \sqrt{2} \leq 1.7 * 4$$

1.7×4=6.8

4.33≤6.8



X_1 = *width of ring wall*

X_2 = *width of base plate* = $2X_1$

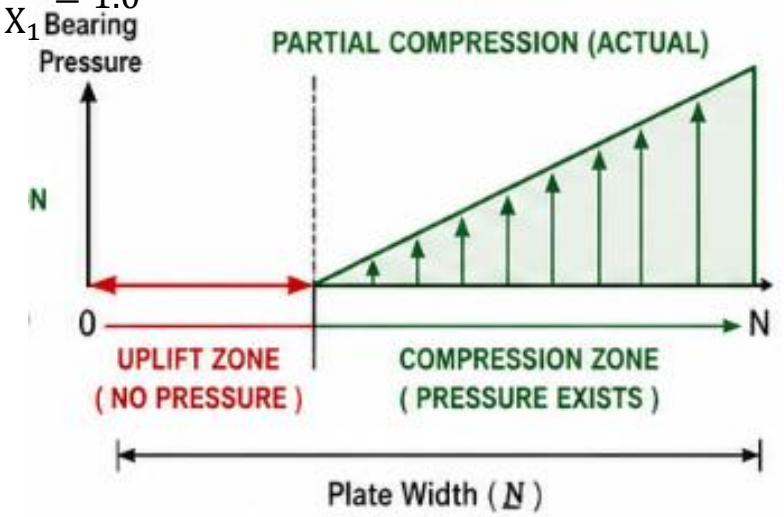
CALCULATE F_p

if $F_p \leq 1.7 f_c'$ o.k.

If increase X_1 → maximum $\frac{X_2}{X_1} = 1.0$

$$F_p \leq 1.7f'_c$$

$$\frac{X_2}{X_1} = 1.0$$



3. Determine Equivalent Eccentricity

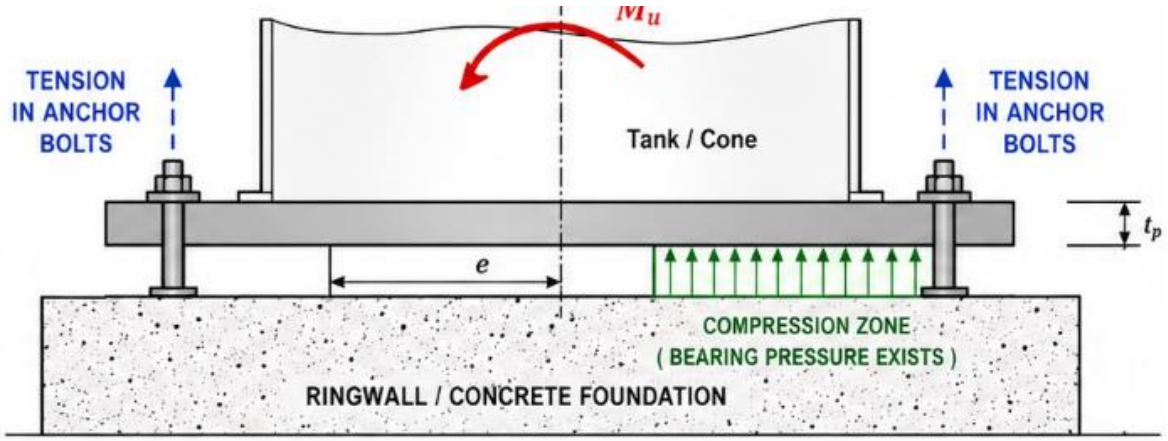
Governing Equation

$$e = \frac{M}{P}$$

Using strip quantities:

$$e = \frac{44.0083}{2.084}$$

$$e = 21.11 \text{ in}$$



3.1. Check Bearing Condition

For full compression:

$e \leq \frac{N}{6}$

Substitute:

$$\frac{N}{6} = \frac{12}{6}$$

$\frac{N}{6} = 2$ in; Since: $\mathbf{21.11 > 2}$

full compression does not occur.

Therefore:

- uplift develops,
- anchor bolts resist tension,
- partial compression exists beneath the annular plate.

$$e = 21.11 \text{ in} > \frac{12}{6} = 2 \text{ in}$$

3.2. Determine Critical Section

Using annular strip geometry:

$$m = \frac{B}{2}$$

Given:

$$B = 12 \text{ in}$$

Thus:

$$m = \frac{12}{2}$$

$$m = 6 \text{ in}$$

$$m = \frac{12}{2} = 6 \text{ in}$$

3.3. Determine Plastic Moment per 1-in Strip

Using the AISC LRFD moment procedure:

Governing Equation

$$t_p = \sqrt{\frac{4M_{plu}}{0.90F_y}}$$

$$t_p = \sqrt{\frac{4M_{plu}}{0.90F_y}}$$

3.4. Determine Strip Plastic Moment

Approximate factored strip moment:

$$M_{plu} = 44.0083 \text{ kip-in/in}$$

3.5. Determine Required Thickness

Substitute:

$$t_p = \sqrt{\frac{4(44.0083)}{0.90(36)}}$$

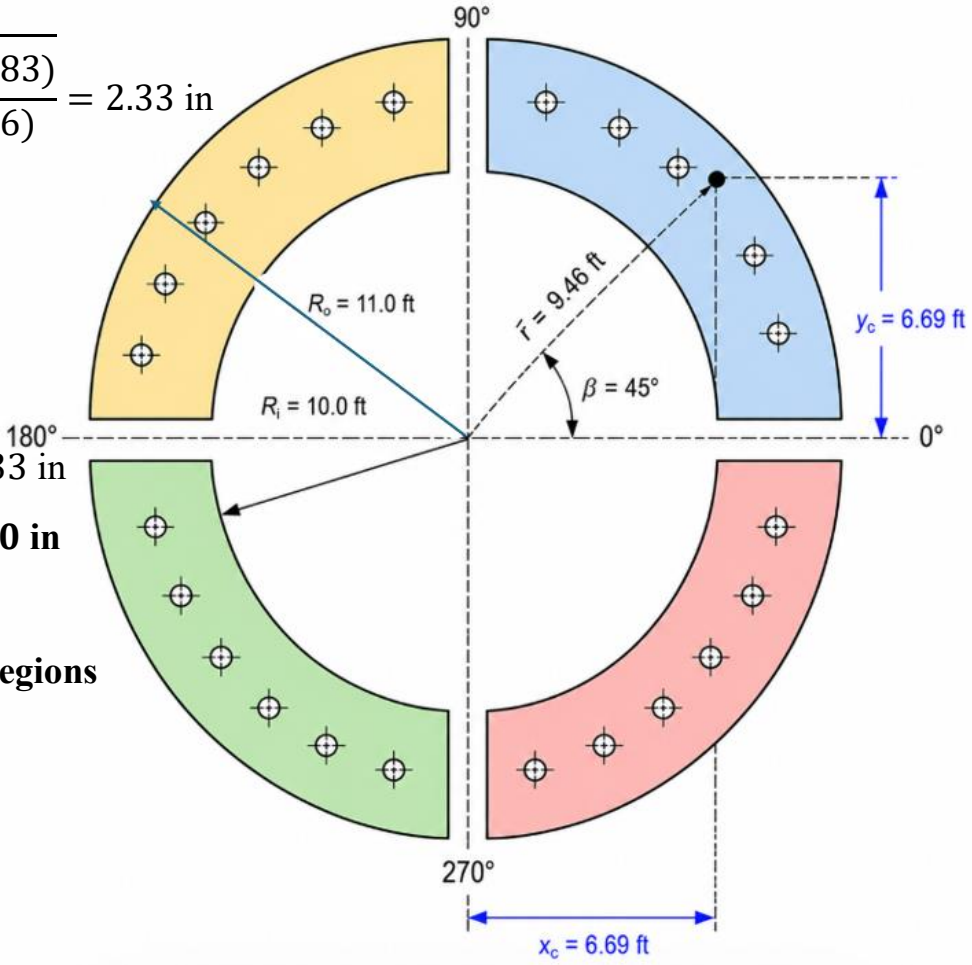
Denominator:

$$0.90(36) = 32.4$$

Thus:

$$t_p = \sqrt{\frac{4(44.0083)}{0.90(36)}} = 2.33 \text{ in}$$

$$t_p = 2.33 \text{ in}$$



3.6. Final Thickness Selection

From moment loading: $t_p = 2.33$ in

Therefore, moment governs. **Adopt** $\boldsymbol{t_p = 3.0}$ **in**

4.0. Local Uplift Bending Between Compression and Uplift Regions

the adopted thickness provides adequate resistance against:

- local uplift strip flexure,
- localized plate deformation,
- annular cantilever bending.

Therefore, local uplift bending is adequate.

5.0. Local Plate Stability / Buckling Check

The compactness ratio of the annular strip is:

$$\frac{B}{t_p} = \frac{12}{3} = 4$$

$$\frac{B}{t_p} = \frac{12}{3} = 4$$

Because the plate is very stocky and compact:

- local plate buckling,
- compression instability,
- local flexural instability,

are not governing limit states.

Therefore, the annular plate is stable against local buckling.

6.0. Segment Angle

Since the annular base plate is divided into four equal segments:

$$\theta_{seg} = \frac{360^\circ}{4}$$

$$\theta_{seg} = 90^\circ$$

Convert to radians:

$$\theta_r = \frac{\pi}{2} = 1.5708 \text{ rad}$$

$$\theta_r = \frac{\pi}{2} = 1.5708 \text{ rad}$$

Plate Thickness

From overturning moment design:

$$t_p = 3.0 \text{ in}$$

Convert to feet: $$t_p = \frac{3}{12} = 0.25 \text{ ft}$$

6.1. Centroid Radius of Annular Segment

Governing Equation

$$\bar{r} = \frac{4\sin(\theta_r/2)}{3\theta_r} \cdot \frac{R_o^3 - R_i^3}{R_o^2 - R_i^2}$$

Using:

$$R_o = 11.0 \text{ ft}$$

$$R_i = 10.0 \text{ ft}$$

$$\theta_r = 1.5708 \text{ rad}$$

Substituting:

$$\bar{r} = \frac{4\sin(1.5708/2)}{3(1.5708)} \cdot \frac{11^3 - 10^3}{11^2 - 10^2}$$

$$= \frac{4\sin(0.7854)}{4.7124} \cdot \frac{1331 - 1000}{121 - 100}$$

$$= \frac{4(0.7071)}{4.7124} \cdot \frac{331}{21}$$

$$= 0.6002(15.7619)$$

$$\boldsymbol{\bar{r} = 9.46 \text{ ft}}$$

6.2 Centroid Coordinates

For Segment 1:

$$\beta = 45^\circ$$

x-Coordinate

$$x_c = \bar{r}\cos\beta$$

Substitute:

$$x_c = 9.46\cos 45^\circ$$

$$x_c = 9.46(0.7071)$$

$$x_c = 6.69 \text{ ft}$$

$$x_c = 9.46\cos 45^\circ = 6.69 \text{ ft}$$

y-Coordinate

$$y_c = \bar{r}\sin\beta$$

Substitute:

$$y_c = 9.46\sin 45^\circ$$

$$y_c = 9.46(0.7071)$$

$$y_c = 6.69 \text{ ft}$$

$$y_c = 9.46\sin 45^\circ = 6.69 \text{ ft}$$

6.3. Segment Centroid Coordinates

Segment	**Angular Range**	**Centroid Angle β**	**$x_c = \bar{r}\cos\beta$ (ft)**	**$y_c = \bar{r}\sin\beta$ (ft)**	**Centroid Coordinates (ft)**
Segment 1	$0^\circ \rightarrow 90^\circ$	45°	+6.69	+6.69	(6.69, 6.69)
Segment 2	$90^\circ \rightarrow 180^\circ$	135°	-6.69	+6.69	(-6.69, 6.69)
Segment 3	$180^\circ \rightarrow 270^\circ$	225°	-6.69	-6.69	(-6.69, -6.69)
Segment 4	$270^\circ \rightarrow 360^\circ$	315°	+6.69	-6.69	(6.69, -6.69)